Improving Hand Anatomy via LoRA Fine-Tuning of PixArt-α

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GitHub: PixArt-Alpha-LoRA **YouTube:** youtu.be/Aoez03xao6E
W&B: WandB logging **Gradio:** Demo

Abstract

Text-to-image diffusion models consistently fail to generate anatomically correct human hands. They usually produce extra fingers, fused digits, and distorted geometry. This is also the problem with models like Pixart-α. We took the problem of hand generation into account and try to solve it by applying Low-Rank Adaptation (LoRA) fine-tuning to PixArt-α on a curated dataset of hand images with detailed and dense captions generated by LLaVA (7B parameters). We explore three dataset configurations combining the 11k Hands and FreiHAND datasets, conduct manual as well as automated (Optuna) hyperparameter tuning, and evaluate checkpoints via FID and T2I-CompBench, and qualitative comparison with Weights & Biases logging. The 11k-only configuration yields the best results due to its cleaner, more consistent image distribution.

1 Base Paper Overview

1.1 Problem Statement

State-of-the-art text-to-image (T2I) models achieve high-quality image generation but require extremely large-scale training, often involving millions of GPU hours, high financial cost, and significant carbon emissions. This makes such models inaccessible to most research groups and limits innovation. There is a need for a compute-efficient framework that can generate high-resolution, photorealistic images while drastically reducing training cost and resource requirements.

1.2 Proposed Solution: PixArt-α

PixArt-α addresses the training efficiency problem through a three-stage decomposition:

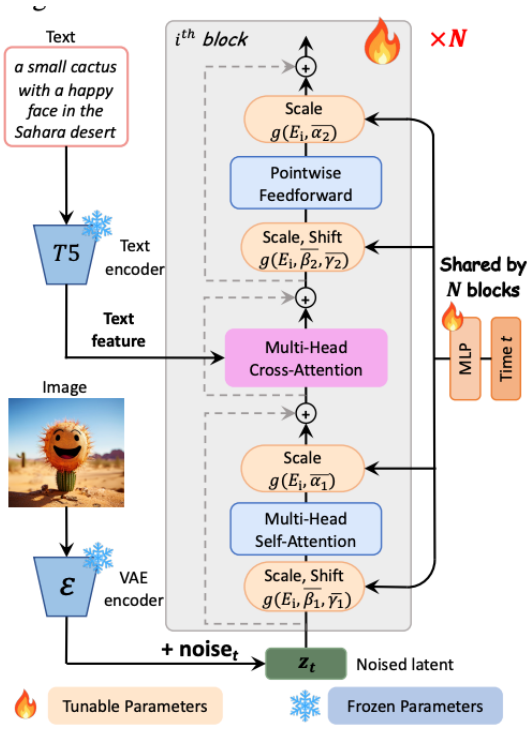
1. **Class-conditioned pixel-level pretraining** on ImageNet to learn low-level pixel statistics without needing expensive text-image pairs.
2. **Concept-alignment fine-tuning** on high-quality text-image pairs to inject language conditioning.
3. **High-resolution aesthetic refinement** at 1024×1024 resolution using human preference data.

Each stage is significantly smaller in scope than training the entire model at ones which learns all these in one go. Learning this correlation in one go requires a lot of compute. If we divide the model training into stages we can get good results in less amount of training.

1.3 Architecture

The backbone of Pixart-α is a Diffusion Transformer (DiT) which replaces the U-Net commonly used in latent diffusion models (such as Stable Diffusion) with a pure transformer architecture that works in the VAE latent space. Key architectural components are:

- **T5-XXL text encoder** (4.6B parameters, frozen) providing rich 4096-dimensional token embeddings for each caption token.
- **Multi-scale VAE** compresses 1024×1024 images into 128×128 latents with 4 channels.
- **DiT blocks** with cross-attention conditioning: each block contains a self-attention sublayer followed by cross-attention over T5 embeddings, enabling fine-grained text alignment.
- **Adaptive LayerNorm** conditioned on timestep and text embeddings for improved training stability.
- **Resolution and aspect-ratio embeddings** injected as additional conditioning signals, enabling flexible output sizes.



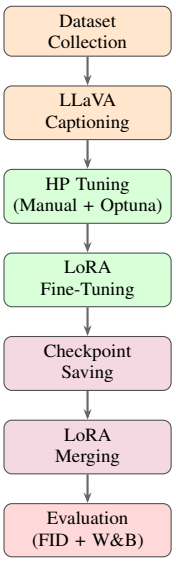
Pixart-α Architecture

1.4 Key Contributions of PixArt-α

- Uses approximately **10%** of the training compute of SDXL, reducing costs from millions of GPU-hours to tens of thousands.
- The modular frozen-encoder design isolates the trainable transformer backbone, making it easy for it to be used in a parameter-efficient fine-tuning.

2 Our Approach

We target the hand-anatomy failure by the model by targeted LoRA fine-tuning of the PixArt-α DiT backbone. Our end-to-end pipeline spans dataset construction, LLaVA-based captioning, LoRA training with extensive hyperparameter exploration, and systematic evaluation. The following figure summarizes the approach.



Pipeline for LoRA fine-tuning of PixArt-α on hand images.

2.1 Dataset Construction

Datasets Used. We experimented with two public hand datasets:

- **11k Hands**: 11,076 images of various views of hands against clean backgrounds. Images are consistently lit, well-framed, and unambiguous which makes it an "easy" training distribution.
- **FreiHAND**: 130,240 images of hand poses captured in the wild with varying backgrounds, occlusion, and object interactions. This is a significantly more challenging distribution due to perspective distortion and partial occlusion.

Dataset Configurations: We trained on three dataset mixes to study the effect of composition:

Config	Composition	Total
Config A	FreiHAND 2000 + 11k 1000	3000
Config B	FreiHAND 500 + 11k 2500	3000
Config C	11k only 3000	3000

All configs use 3000 total images to keep compute constant across experiments and isolate the effect of dataset composition.

LLaVA Captioning: Raw datasets often lack the fine-grained precise descriptions required to guide the model toward correct hand generation. To address this, we use dense captions generated using LLaVA. Each image in the dataset involving hands is passed through LLaVA with the following prompt: "Describe this hand image in detail: how many fingers are visible, their pose and separation, knuckle visibility, palm orientation, skin texture, and lighting conditions."

This prompt helps the model to focus on features such as finger count, pose, and texture. The generated captions are stored in a structured JSON format.

During evaluation, we apply another prompt to generate captions for images from the COCO dataset: "Describe this image and its style in a very detailed manner."

2.2 LoRA Fine-Tuning

Instead of retraining the entire PixArt-α model, we use Low-Rank Adaptation (LoRA) to efficiently adapt it for better hand generation.

The key idea is to slightly adjust the model's existing weights by learning small, low-dimensional updates rather than modifying the full weight matrices. These updates are added on top of the original weights, allowing the model to learn new features (such as correct hand anatomy) without forgetting its original knowledge.

LoRA is applied to the attention layers of the Diffusion Transformer, specifically the query, key, value, and output projections. The rest of the model, including the VAE and T5 text encoder, is kept completely frozen. This significantly reduces the number of trainable parameters.

At the start of training, the LoRA updates are initialized such that they do not affect the original model's behavior. As training progresses, they gradually learn corrections that improve hand structure, such as proper finger count and separation.

With a small rank (e.g., $r = 16$), LoRA introduces only a tiny fraction ($\sim$0.3%) of additional parameters. This makes fine-tuning feasible on a single GPU with limited memory, while still achieving significant improvements.

Training follows the standard diffusion objective, where the model learns to remove noise from images step by step, guided by the input text captions.

2.3 Hyperparameter Tuning

Automated Search with Optuna We initially experimented with Optuna for automated hyperparameter tuning. It helped us understand good ranges for parameters like learning rate,

dropout, scheduler, and gradient accumulation. However, we did not directly use the exact values suggested by Optuna.

We observed issues with LoRA rank and alpha. Optuna often suggested combinations that did not follow the common rule ($\alpha \approx 2 \times r$), for example ($r = 4, \alpha = 64$). Such values led to unstable training and poor output quality. Since α controls how strong the LoRA updates are, incorrect values can either make the model unstable or too weak to learn.

We also noticed that Optuna suggested very precise values (e.g., random decimal learning rates). Instead of using these directly, we used Optuna only to find approximate ranges, and then manually selected clean, stable values. **Final Strategy:**

- LoRA rank and alpha were not tuned using Optuna and were set manually.
- We followed the rule $\alpha = 2 \times r$ and chose $r = 16$.
- Other parameters were chosen based on Optuna ranges and manual experimentation.

Final hyperparameter configuration used for training.

Hyperparameter	**Value**
Learning Rate	5×10^{-6}
LoRA Rank (r)	16
LoRA Alpha (α)	32
Batch Size	2
Epochs	15
Dropout	0.1
LR Scheduler	Cosine
Warmup Steps	200
Gradient Accumulation	4
Resolution	512

In summary, Optuna was mainly used to understand good value ranges, while the final hyperparameters were selected through manual tuning.

2.4 Evaluation Infrastructure

All training runs were documented with Weights & Biases (W&B) logging (see W&B runs), which tracked: training loss, gradient norms, LoRA weight norms, and generated image grids every 100 steps. After training, a dedicated evaluation script iterated over all saved checkpoints (every 500 steps), generating images for a fixed set of 10 hand-related prompts, and logging all outputs to W&B for side-by-side visual comparison. The checkpoint at step 500, the final checkpoint before merging, and the merged model (with LoRA weights folded into the base transformer) were all evaluated as standalone pipelines.

3 Experiment Results and Visual Outputs

3.1 Dataset Curation

		COCO Dataset	Custom Dataset
Original Model	PIXART-α	43.282	126.734
LoRA FT Model	Checkpoint-500	47.876	116.406
	Checkpoint-final	46.413	109.793
	Merged Model	46.287	109.607

FID comparison across models and datasets.

Why 11k outperforms FreiHAND mixes. The FreiHAND dataset, while large, contains views with heavy perspective distortion, cluttered backgrounds, and partial occlusion. These characteristics are difficult for LoRA adapters to learn alongside clean anatomy (which was the main focus in our case). The 11k Hands dataset, by contrast, is a controlled, consistent distribution: hand views against uniform backgrounds, uniformly illuminated. This simpler distribution allows the LoRA adapter to focus its limited rank budget on features such as finger count, separation etc. rather than on background or viewpoint variation. Increasing the proportion of 11k images in the training mix (Config A $\rightarrow$ B $\rightarrow$ C) consistently improved the images semantics visually.

3.2 T2I-CompBench Evaluation

We also evaluate on T2I-CompBench, a benchmark for compositional text-to-image generation. The following table reports scores for attribute binding, spatial reasoning, non-spatial reasoning, and complex compositionality.

		Attribute Binding	Spatial	Non-Spatial	Complex
Original Model	PIXART-α	0.482	0.039	0.301	0.351
LoRA Fine-Tuned Model	Checkpoint-500	0.476	0.041	0.338	0.348
	Checkpoint-final	0.473	0.026	0.319	0.362
	Merged Model	0.490	0.057	0.325	0.339

T2I compbench comparison across models

3.3 Visual Comparison

Figure 1 shows a qualitative comparison across four model variants (base, checkpoint-500, checkpoint-final, and pixart-merged) on diverse COCO prompts, while Figure 2 focuses specifically on hand generation quality.

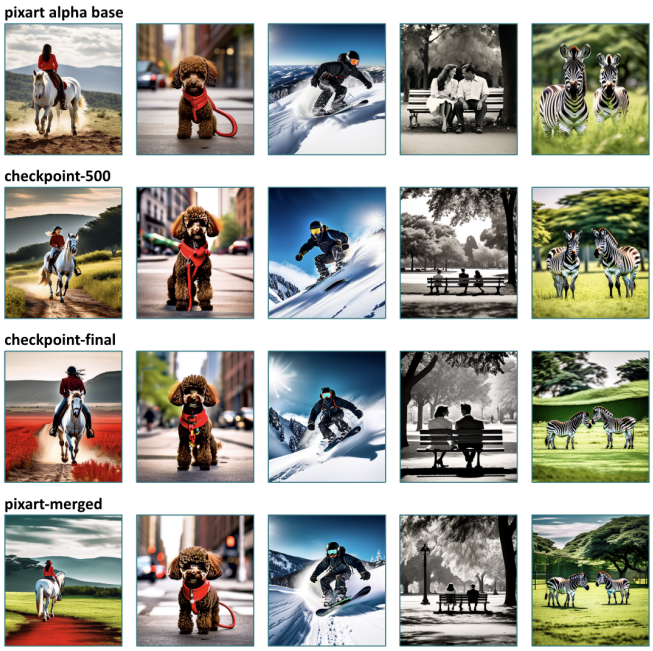


Figure 1: Visual comparison of base, LoRA checkpoint, and merged model outputs across diverse COCO prompts (horse riding, dog on leash, snowboarder, park bench, zebras).

Figure 2: Visual comparison of base, LoRA checkpoint, and merged model on prompt involving hand genration

The base model produces anatomically accurate results on general COCO prompts but consistently generates fused fingers or entirely malformed hands. LoRA fine-tuning improves both domains: checkpoint-500 already shows sharper subject rendering on COCO scenes and visible finger separation in hand samples. The checkpoint-final and pixart-merged variants maintain COCO quality while continuing to improve hand anatomy. The merged model achieves the best overall results across all prompts tested.

References

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